

## ORGANISM AND PROCESS PARAMETERS

strain composition (C, N, P) · productivity · operating days · reactor/pond type  
harvesting recovery · drying · extraction · plant scale

**SHARED MASS AND ENERGY BALANCE**  
**flows per kg of product + annual production**

*computed ONCE; both layers read it*

### TEA LAYER

prices · CAPEX factors · labour · installation and indirect factors · depreciation · discount rate · tax

### LCA LAYER

characterization factors · grid mix  
biogenic-carbon convention · system boundary

production cost · CAPEX  
NPV · IRR · MEPP

gross GWP · biogenic adjustment · net GWP  
CED · water · land · eutrophication

*prices and characterization factors are NOT inputs to the physical balance*